VIEWS

Delivery Note

Darts Deep-Learning Models
Calibration Forecasts, Global PRIO-GRID Month

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PRIO – Oslo

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1 What These Models Are

All nine are **off-the-shelf deep-learning models for time series**, taken from the Python library **Darts** (`unit8co/darts`). The architectures are the library's, unmodified — each is a published model used as Darts implements it. What we supply is the data, the configuration, and for six of the nine a loss function of our own (Section 4).

They are *global* models in the forecasting sense: one model is fitted across all grid cells at once, rather than one model per cell.

Table 1: Scope of the delivery.

Data backend	The dataFactory, for all nine models
Spatial coverage	Global land — all 64,818 PRIO-GRID land cells
Temporal coverage	The **calibration** partition: trained on January 1990 – December 2017, forecasting **January 2018 – December 2021** (`month_id` 457–504)
Outcomes	State-based, non-state and one-sided fatalities (`lr_ged_sb`, `lr_ged_ns`, `lr_ged_os`)

2 The Models

Table 2: The nine delivered models.

Model	Type	Loss	Predictors
`dark_river`	N-BEATS	MSE	Outcomes only
`old_rules`	N-BEATS	Spotlight	Outcomes only
`blue_ocean`	N-BEATS	Spotlight	+ 69 covariates
`silent_fox`	TSMixer	MSE	Outcomes only
`dancing_monkey`	TSMixer	Spotlight	Outcomes only
`brave_heart`	TSMixer	Spotlight	+ 69 covariates
`red_hawk`	TiDE	MSE	Outcomes only
`dark_necessities`	TiDE	Spotlight	Outcomes only
`golden_eagle`	TiDE	Spotlight	+ 69 covariates

The names are arbitrary labels and carry no meaning.

Within each block of three, exactly two settings vary — the loss function, and whether the model sees covariates. Everything else is identical: layer widths, depth, training epochs, learning rate, early stopping. So inside a block, each row differs from the one above it in a single setting. Across blocks the architecture changes and the comparison is a different one.

- **Outcomes only** — the model's inputs are the past values of the three outcomes and nothing else. Purely autoregressive. Six models: the first two rows of each block.

- **+ 69 covariates** — the same three histories plus 69 predictors: ACLED event and fatality counts, land cover shares, population and built-up area, infant and child mortality, V-Dem governance indices, SHDI development indices, travel time to cities, mineral and petroleum deposits, growing season and harvested area, and terrain. Three models: the third row of each block.

Note that the two splits are not the same split. Six models are outcomes-only and six use the Spotlight loss, but they are different sixes: `blue_ocean`, `brave_heart` and `golden_eagle` have covariates *and* Spotlight, while `dark_river`, `silent_fox` and `red_hawk` have neither.

There is no model combining MSE with the covariates. Three configurations exist, not four.

3 The Model Types

All three are feed-forward neural networks built from ordinary dense layers. None uses recurrence, convolution or attention.

N-BEATS

A deep stack of fully connected layers with backward and forward residual links. It builds the forecast as a sum of learned basis functions, which is what makes it partly interpretable. Designed for univariate point forecasting and benchmarked on the M3, M4 and TOURISM competition datasets.

> Darts: `NBEATSModel`
> Paper cited there: Oreshkin, B. N., Carpov, D., Chapados, N. & Bengio, Y. (2020), *N-BEATS: Neural basis expansion analysis for interpretable time series forecasting*, ICLR 2020. arXiv:1905.10437

TSMixer

Dense layers only, applied alternately along the time axis and across the predictor variables, so the model mixes temporal structure and cross-variable structure in turn. Built for the multivariate case.

> Darts: `TSMixerModel`
> Paper cited there: Chen, S.-A., Li, C.-L., Yoder, N., Arik, S. O. & Pfister, T. (2023), *TSMixer: An All-MLP Architecture for Time Series Forecasting*, Transactions on Machine Learning Research. arXiv:2303.06053

TiDE

An encoder–decoder, again dense layers only. It compresses the history and the predictors into a single code, then decodes the whole 36-month path from it in one pass.

> Darts: `TiDEModel`
> Paper cited there: Das, A., Kong, W., Leach, A., Mathur, S., Sen, R. & Yu, R. (2023),

Long-term Forecasting with TiDE: Time-series Dense Encoder. arXiv:2304.08424

One Setting All Nine Share

All nine use **reversible instance normalisation**: each cell's series is standardised before the network sees it and the transformation is inverted on the way out. The model therefore fits on a per-cell normalised scale, while the forecasts you have are in counts.

> Paper cited by Darts for all three models: Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.-H. & Choo, J. (2022), *Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift*, ICLR 2022.

4 The Loss Functions

MSE — ordinary squared error, every cell-month weighted equally. The library's own implementation, a thin wrapper around PyTorch's. Used by `dark_river`, `silent_fox` and `red_hawk`.

Spotlight (`SpotlightLossLogcosh`) — ours, not off the shelf. A three-part objective computed after an `asinh` transform. It scores the *shape* of the error only in cells where the observation or the forecast exceeds one fatality; separately penalises the *level* in cells whose observation exceeds one fatality; and separately penalises forecasting conflict in cells that stayed below it. It uses `log cosh` rather than squared error, so a single very large observation does not dominate the gradient. Used by the other six.

What You Should Know About Spotlight

It is Dylan's own construction, developed since March 2026; the variant used here dates from April and is the fifth iteration of a family of related objectives. He has written a detailed account of how it works. One of its two parameters is derived rather than tuned — the threshold separating "event" from "quiet" cells is $\text{asinh}(1) \approx 0.88$, exactly one fatality on the transformed scale. The other, a weight on a spectral term, was chosen by hyperparameter sweep.

What does not exist is any verification of it.

- **No citation.** I searched the source files, their documentation, the commit history and the architecture decision records. Two of the other objectives in that same in-house library do carry references to published papers, so this is not a codebase that never cites anything.
- **No tests.** The library's test suite covers eight of its objectives and this is not among them. Nothing checks its gradients, nothing compares it against a reference implementation, and nothing establishes that the code computes what the written account says it computes.

That verification belongs with the author in the first instance. Tests pinning the gradients and a derivation someone else can follow line by line are what make a bespoke objective reviewable; absent those, an outside review means reconstructing the intended design before you can judge the built one. I have not spent my time that way, and I think the order matters: tests and

derivation first, review after. So I am not in a position to vouch for this objective, and I would rather say so than imply an assurance I have not earned.

In practice: it could be excellent, adequate, or quietly wrong, and nothing available to me distinguishes those three. That bears on how much weight a comparison between the Spotlight models and the three MSE ones can carry.

Dylan can explain the design. If the evaluation turns on it, the next step is tests and a written derivation from him, after which someone with the relevant background can review it.

5 What Is in the Files

One folder per model. Inside each: thirteen Parquet files, `..._00` through `..._12`, and a short text file recording when the model ran, which version of the software produced it, and which configuration it used.

The thirteen files are thirteen forecast origins, stepping forward one month at a time. An origin is the last month the model was allowed to see; from there it forecasts the next 36 months.

Table 3: The thirteen forecast origins.

File	Last month seen	Months forecast
`..._00`	December 2017	January 2018 – December 2020
`..._01`	January 2018	February 2018 – January 2021
	...	
`..._12`	December 2018	January 2019 – December 2021

Within every file the earliest month is horizon 1 and the latest horizon 36, so a given target month sits at a different horizon in each file containing it. Metrics computed per origin keep those apart; metrics pooled across origins mix them.

Each file holds 2,333,448 rows — 36 months × 64,818 cells — and five columns.

Table 4: Columns in every file.

Column	Contents
`month_id`	Month
`priogrid_id`	Grid cell
`pred_lr_ged_sb`	Forecast state-based fatalities
`pred_lr_ged_ns`	Forecast non-state fatalities
`pred_lr_ged_os`	Forecast one-sided fatalities

Units are fatality counts, untransformed — not logged, not scaled. A value of 2.4 means 2.4 fatalities. Values are continuous, not integers.

Nine models × thirteen origins = **117 files, about 2 GB** in total.

6 Handing It On

Observed Values Are Not in This Folder, and Do Not Need to Be

The evaluation joins forecasts to observed fatalities on `month_id` and `priogrid_id`. Those observations are not included here, and Sonja's ensemble-construction tool does not need them to be: it fetches actuals itself from a viewser queryset named in its own configuration, and also accepts an actuals dataframe passed in directly. They can equally be pulled from the dataFactory.

Three things about that tool are worth flagging to whoever runs it, taken from reading its current source rather than from memory.

- **The file naming already matches.** It expects `predictions_calibration_{timestamp}_{ts}.parquet` with `ts` running `00`–`12`, which is exactly what is here.
- **Its shipped default configuration is country-month, not PRIO-GRID.** The default keys are `(country_id, month_id)` and the default actuals queryset is a country-level one. The evaluation code itself is indifferent to the level — it merges on whatever keys the configuration names — so pointing it at these models is a configuration change, to `(priogrid_id, month_id)` and a PRIO-GRID actuals queryset, not a code change.
- **It resolves prediction files inside the repository layout**, at `views-models/models/<model>/data/generated/`, rather than from an arbitrary folder. The files here are laid out one folder per model and will need placing accordingly, or the paths adjusting.

If a Research Assistant Runs It Instead

The instruction is: *iterate over the thirteen files per model, join each to the observed values on* `month_id` *and* `priogrid_id`*, and compute point metrics per origin so that horizon effects stay separable.* Parquet reads in one call — `pandas.read_parquet` in Python, `arrow::read_parquet` in R.

Two things to tell them explicitly, because both will otherwise be guessed wrong:

1. forecasts are counts, not logs — no back-transformation;
2. there is no distribution — no distributional scores.

7 What Is Not Included

Table 5: Not included in this delivery.

Item	Where it is instead, or why not
Observed values	Fetched by the ensemble tool, or from the dataFactory (Section 6)
Evaluation metrics	None computed — see Section 8
Training logs	Not shipped. `calibration_log.txt`, which the ensemble tool can optionally read to resolve run timestamps, is among them; timestamps can be passed explicitly instead
Other partitions	Calibration only — no validation partition, no true forecasts
Model artefacts	No weights, no checkpoints

8 Additional Notes

No model here has a predictive distribution. These are point forecasts — one number per cell-month, no draws, no quantiles, no intervals. All nine are configured as deterministic predictors, so a different export would not recover a distribution; it was never produced. Nothing that scores a *single model's* uncertainty — CRPS of its predictive density, interval coverage, density calibration — can be computed from these files.

One qualification, since it would otherwise look like a contradiction in the output: the ensemble tool can still produce a CRPS. It does so by treating the member models' point forecasts as an empirical sample and scoring the spread across models. That is a property of the ensemble, not of any model in it, and it is not the same quantity as a CRPS computed from a model's own posterior. Worth knowing before the two are compared.

Nothing here has been evaluated. I checked only that the files are not broken: no negatives, no missing values, no duplicate cell-months, the expected row counts, and totals within the same order of magnitude as observed. That rules out a failed back-transformation or a collapsed model and establishes nothing else.

It is nine models, not eleven. Two others were set up to draw 100 samples per forecast, making them probabilistic counterparts of `dancing_monkey` and `dark_necessities`. A point delivery needs one value per cell, and at one sample their configurations are byte-identical to those siblings — the same model twice. Running them as intended is blocked by memory: 100 samples for 64,818 cells across thirteen origins needs far more RAM than any machine we have.

One observation will dominate any squared-error metric. December 2021, cell 148759 has **113,395** observed fatalities; the next largest anywhere in the window is 5,395. That month falls in only one of the thirteen origins (`..._12`), where that single cell is about 41% of observed fatalities. I do not know whether it is a real event or a data error. Either way it is better settled before the numbers are produced than explained afterwards.

Column naming matches the tool. These files use `pred_lr_ged_sb`, which is the ensemble tool's own default `target_column`. The September delivery used `pred_lr_sb_best` instead, so

if anything needs renaming when the two are compared it is that one, not this. Renaming either takes minutes and needs no re-run.

Comparison with the September delivery. You were sent three variants per model then — a mean, a 95th percentile and a conditional mean — each computed from that model family's predictive distribution. These nine have no distribution, so only the mean has an equivalent here. If the evaluation work has been built on one of the other two, the two deliveries are not like-for-like. A correction while I am here: my September note said the full distributions had been retained so alternative summaries could be produced without re-running. They were not kept.